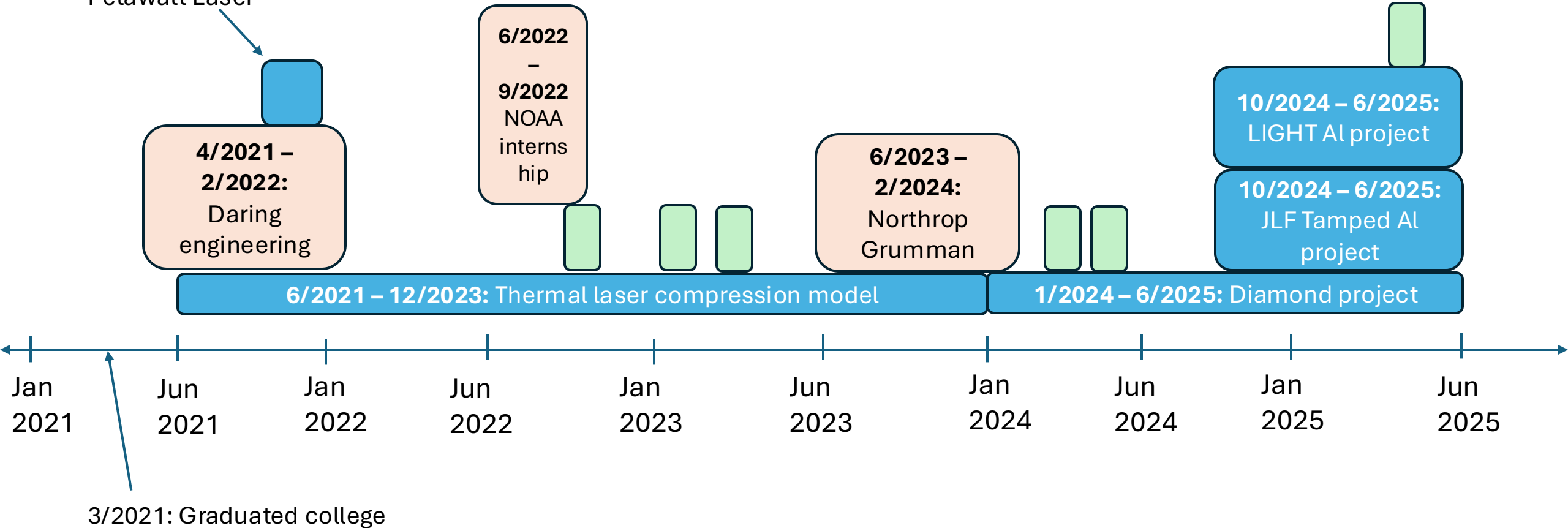
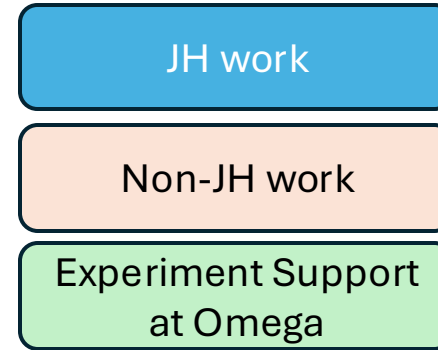


Timeline

11/2021 – 12/2022:
Experiment
support at Texas
Petawatt Laser



JLF Experiment

Project Requirements

- Janus laser at Lawrence Livermore National Labs' Jupiter Laser Facility
- Conduct scan of tamped materials at laser intensities from 10^9 - 10^{11} W/cm², varying: pulse duration, tamper & sample material and adhesion method
- 5-week experiment, 4-person team

Outcomes

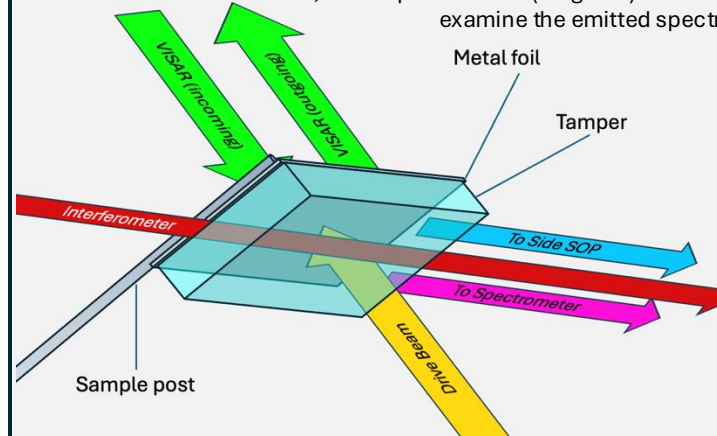
- Successfully collected data on 134 laser shots.
- Analysis underway

My contributions

- I was responsible for VISAR, an optical diagnostic to get the velocity of the back surface (alignment + live analysis)
- Live analysis of VISAR determined diagnostic parameters, which optimized data quality
- simulated experiment
- Prepared samples

Setup & Diagnostics

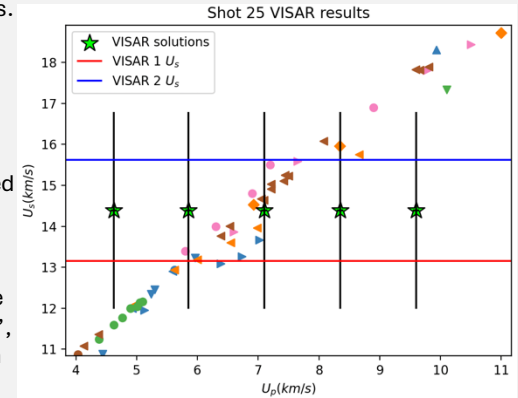
Below – Setup for this expt. The drive (yellow) passes through the transparent tamper and ablates the metal foil. An interferometer (red) is used to view the plasma expansion. VISAR (green, my responsibility) is used to get the velocity of the rear surface of the foil. A streaked optical pyrometer (blue) looks at the sample from the side to view ablation, and a spectrometer (magenta) is used to examine the emitted spectra.



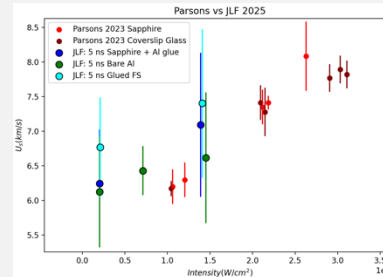
Live Analysis – Determination of Etalon Change

Right – VISAR results for one of our shots.

Measured U_s (vertical axis) is compared to U_p solutions from the analysis. This analysis determined the spacing of surface velocity solutions is too close (see how one data point is “best”, but 3 could explain the data). This dictated the use of smaller etalons.

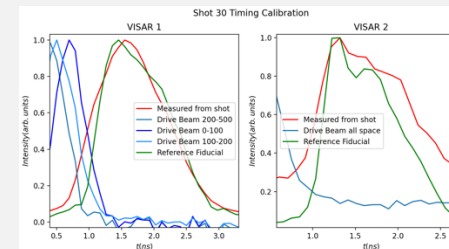


Live Analysis – Comparing Shock Velocities



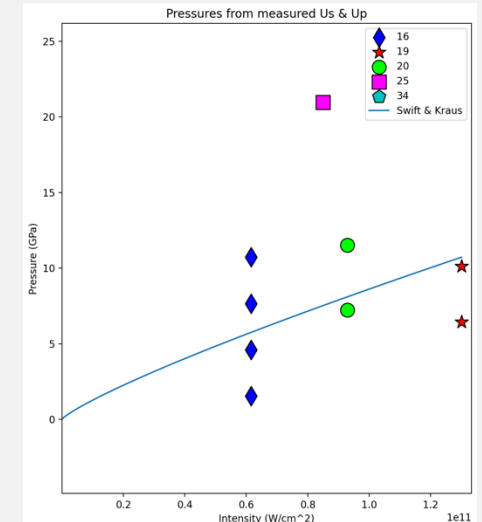
Left – Comparing rough estimates from live analysis to published data. Our data showed good agreement with published data.

Right – Comparing timing fiducial alignments of the 2 VISARs



Live Analysis – Measured v Expected Pressures

Right – results from experiment compared to predicted ablation pressures from Swift & Krause. This data showed good agreement for the infrared drive laser (shots 16, 19 & 20) but bad agreement for the green drive laser. This analysis demonstrated a timing error in the green laser and we recalibrated the timing



Diamond Shock

Project Requirements

- Former Grad student Gin Li collected shock data on Diamond to get the 1st in-situ XRD measurement of the principle hugoniot
- Li left before resolving the data
- I was given the data to analyze and resolve

My Contributions

- Completed XRD and VISAR (velocimetry analysis to get pressure) for the dataset (9 datapoints, 6 usable)
- Compared my measurements to other work and showed a spread in densities which was much too large
- Developed code to simulate rotation and compression in two different ways (Unit cell representation & Ewald's sphere)

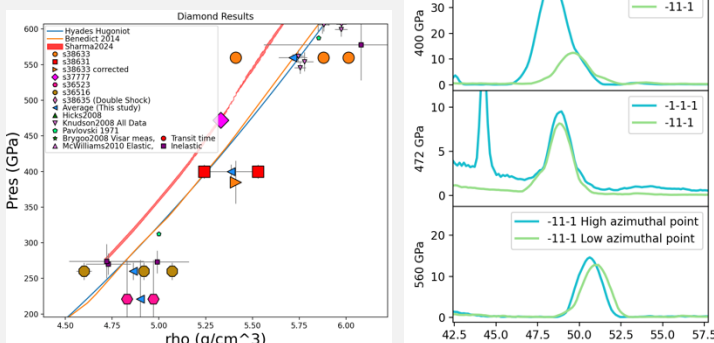
Outcomes

My optimized Ewald's sphere rotation code found solutions to explain all data. There is a rotation of ~ 10 degrees from the elastic wave

Initial Data Analysis

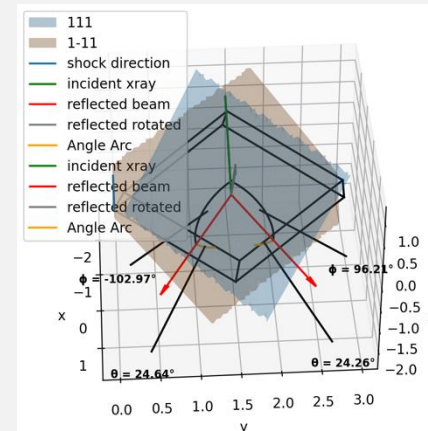
Right: Lineouts from diffraction peaks @ different pressures. Data at lower pressures shows the peaks clearly have different location and intensity. The two highest pressures show the same location and intensity

Below: Hugoniot results from using measured peaks. Different peak locations give high variance in density.



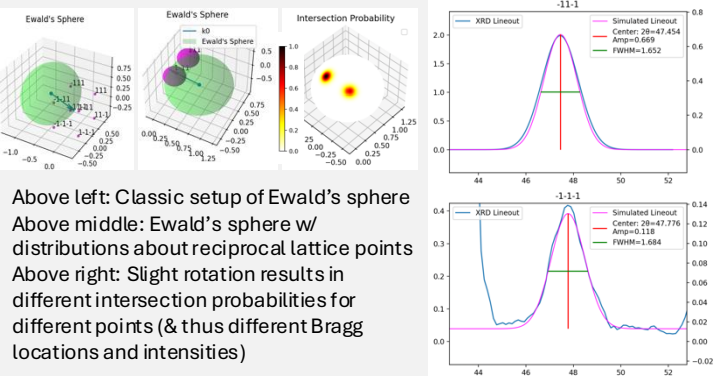
Simulating non-uniform compression

- The first attempt to explain the data involved some non-uniform compression of the unit cell
- I compiled a substantial codebase to simulate and visualize XRD data under different compressions of the diamond unit cell
- I optimized to measured parameters using a variety of techniques but did not achieve any solutions.
- The results of one of the optimizations are at right. My code takes a rotation of the unit cell and compressibility along the shock direction.

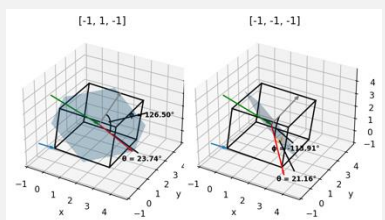


New unit cell parameters are then determined using the inputs and keeping density constant.

Simulating Rotation



Above left: Classic setup of Ewald's sphere
Above middle: Ewald's sphere w/ distributions about reciprocal lattice points
Above right: Slight rotation results in different intersection probabilities for different points (& thus different Bragg locations and intensities)



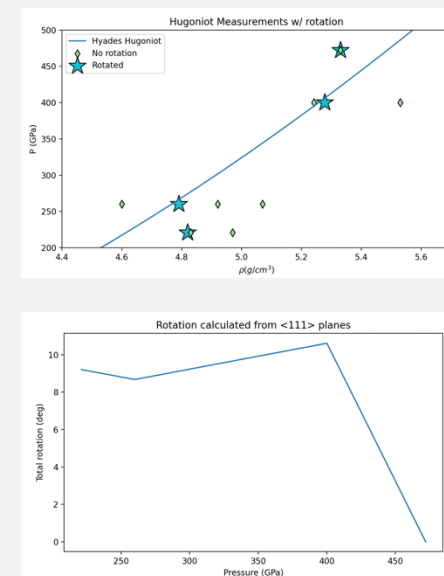
Above: Rotation that satisfies the different diffraction peaks for a certain shot
Left: Diffraction angles from the unit cell from these planes for this rotation

Results w/ rotation

I optimized the rotation for each data point to match the measured diffraction peaks

Top: Results from optimized rotations show good agreement with the hugoniot. Variation in densities is resolved, and data is publishable.

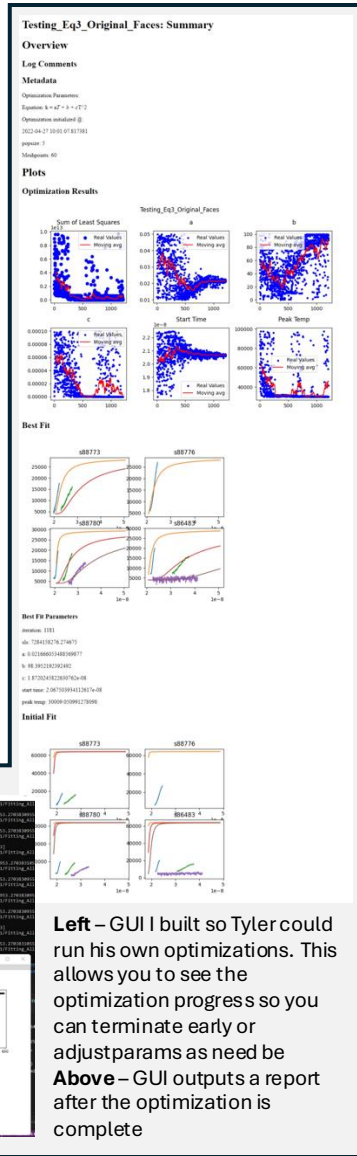
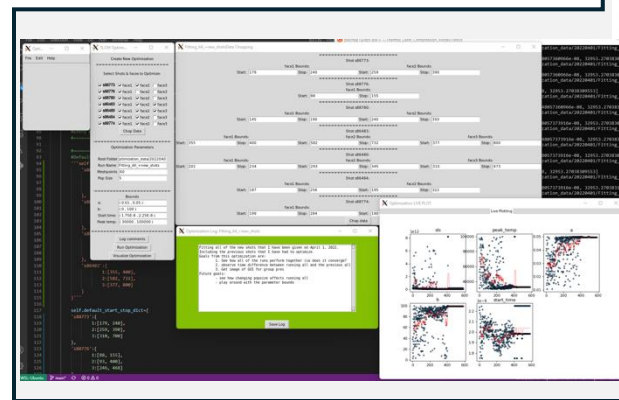
Bottom: Rotation from each optimized datapoint. Rotation is constant (~ 10 degrees) below the Hugoniot Elastic limit, and none above the HEL



Thermal Laser Compression Model

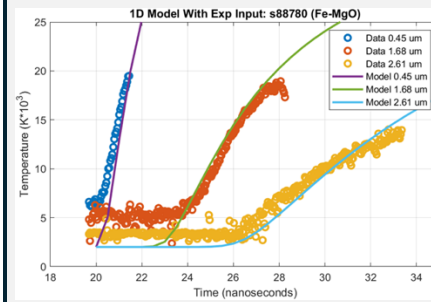
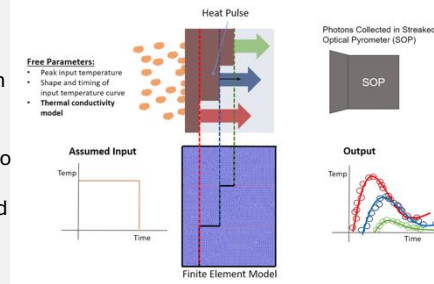
Requirements, Contributions & Outcomes

- Built complex FEM model for high temp thermal conductivity experiments
- Experimental data was the project of a grad student. I did all model and coding work
- After developing a robust codebase, I built a GUI and optimization platform so the grad student could run his own simulations



Plan of Analysis + Matlab Model

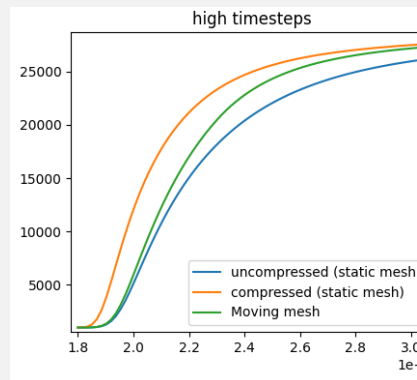
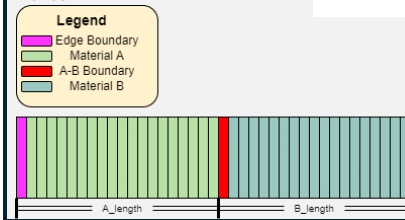
Right – layout of problem. A gold halfraum is ablated to give a smooth heat pulse on a stepped Fe sample with minimal pressure increase. An MgO window was used to constrain the Fe. The thermal conductivity should relate the temperatures at different step lengths



Left – Lineout from base thermal conductivity FEM in Matlab. This model did not perform well on other data. It looks pretty good on this data, but the thermal conductivity is was too high. We needed some way to let the thermal conductivity change with temperature.

3rd iteration – Moving Mesh

Although there is minimal pressure increase in the sample, there is still some compression happening. This causes the measurement location to change as the sample compresses. The adjusted code took in a pressure wave and shrunk each cell at each timestep accordingly. MgO and Fe have different compressibilities and therefore shrink at different rates.

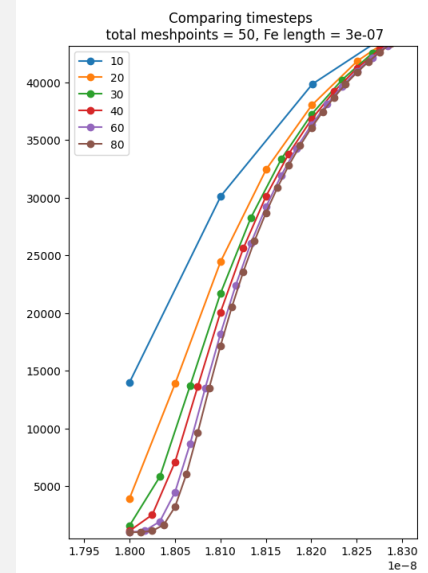


Above – moving mesh (green line) goes from the uncompressed state at early times to the compressed state at later times

Left – Pictorial description of this FEM w/ boundary

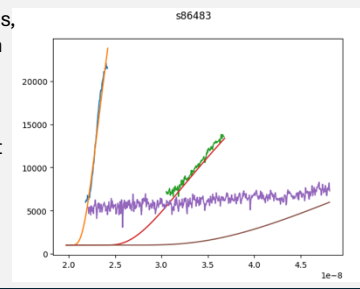
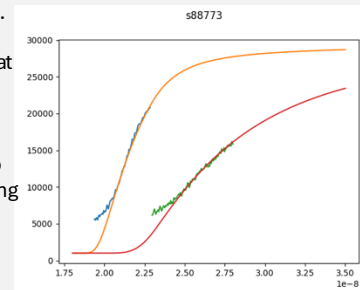
2nd iteration – Python model

- right: Determination of number of timesteps required. Fewer timesteps (10 & 20) clearly leads to an inaccuracy in the beginning of the trace. Above 60, the difference is quite small and negligible
- This model was built in python (with fenics) instead of Matlab to allow for custom thermal conductivity equations as a function of temperature and pressure
- Python also allowed for infinite tweaking of the model and rapid optimizations. This was useful as we iterated the model to match the data.



Results

Right – Final results of this work. We got a good match to the temperature data. MgO is optically transparent at room conditions, but transitions to opaque at high temperature and is therefore emitting photons. This is not well understood and was catastrophic for this project. This is why I was forced to cut off the data early (only the beginning of the trace is visible). With such little data and so many free parameters (conductivity equation, parameter values for conductivity equation, compressibility, number of meshpoints, number of timesteps, etc), there was a multiplicity of solutions.



Other features added to the model were: square heat input v custom heat input, user-defined thermal conductivity equations, extensive plotting features, standardized data format for reading in expt data, GUI for monitoring of optimization progress

LIGHT Experiments

Project Requirements

- LIGHT is a new laser commissioned in 2024 (the year before this work)
- MSEE group (which funded me) had developed uncertain hugoniot
- LIGHT facility would like to calibrate their SOP (temp instrument) at high pressures
 - Csl is a known high-pressure temperature calibrant

My Contributions

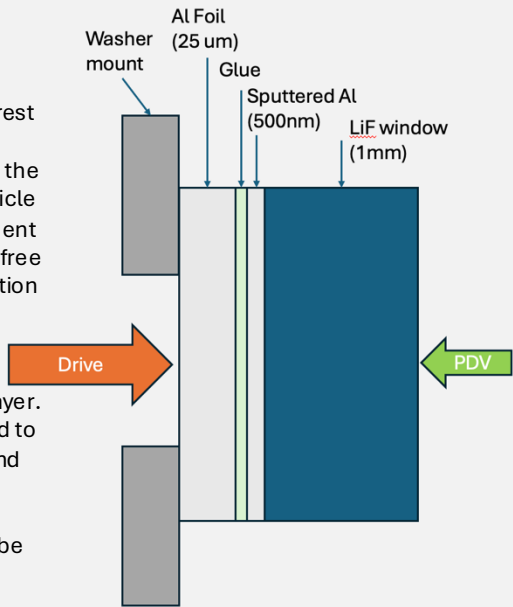
- Designed experiments on Al to 1. simplify SOP temperature calibration with Csl and 2. collect data for uncertain EOS
- Ordered and prepared samples
- Directed shots

Outcomes

- Successfully collected data for uncertain EOS. Analysis underway
- Pressure too low for Csl calibration. Discussing methods to increase pressure

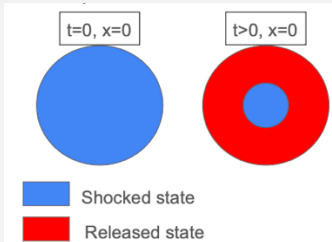
Sample Setup

The sample of interest here is Al. The LiF window constrains the Al to make the particle velocity measurement more accurate (no free surface approximation required). Al is sputtered onto the LiF to minimize the effect of the glue layer. 25um foil was glued to the sputtered Al, and the entire package was mounted on a washer in a lens tube for the experiment



Determination of Sample Dimensions

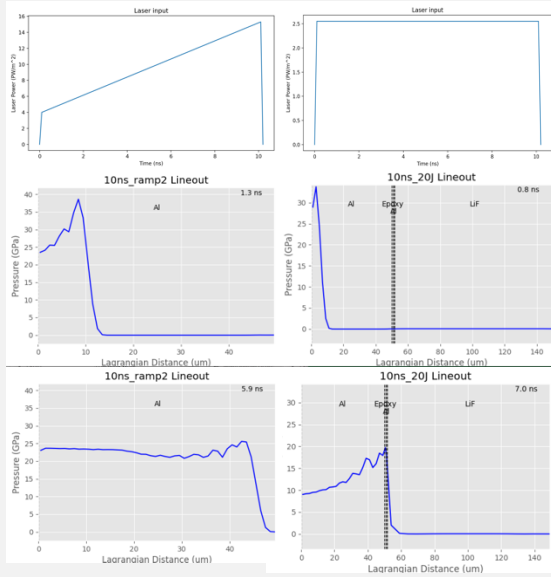
- We had to design for the eventual Csl experiment
- In order to use the velocity from the back side of the sample to get pressure, we need to use the hugoniot relations
- Pressure release wave cannot catch up before the shock reaches the rear side of the sample



Max Csl length (um)				
Pulse time (ns)	R=3 (optimistic)	R=4	R=5 (conservative)	
0.2	167.626667	125.96	100.96	
0.4	168.586667	126.92	101.92	
0.6	169.546667	127.88	102.88	
0.8	170.506667	128.84	103.84	
1	171.466667	129.8	104.8	
1.2	172.426667	130.76	105.76	
1.4	173.386667	131.72	106.72	
1.6	174.346667	132.68	107.68	
1.8	175.306667	133.64	108.64	
2	176.266667	134.6	109.6	

Simulations

I simulated different laser pulses to determine laser inputs required to achieve desired pressure



Primary Analysis

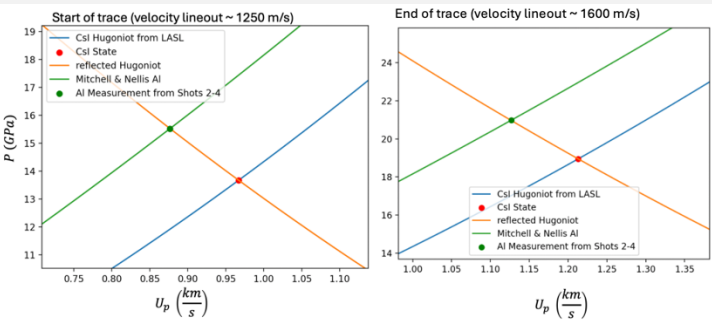
We received a velocity lineout from the PDV instrument, which corresponded with a range of velocities. I applied the correction at right and did impedance matching calculations (below) to show this pressure is too low for our Csl calibration (we need at least 25 GPa)

Jensen 2007
$$v^* = n_0 D - n(D - v)$$

Jensen + Mitchell & Nellis

$$v = \frac{v^* - 5.386(n_0 - n)}{1.339n_0 - 0.339n}$$

apparent velocity	n_0	n	real velocity
1.25	1.4046	1.4015	0.87738954
1.4	1.4046	1.4015	0.98410167
1.6	1.4046	1.4015	1.12638451



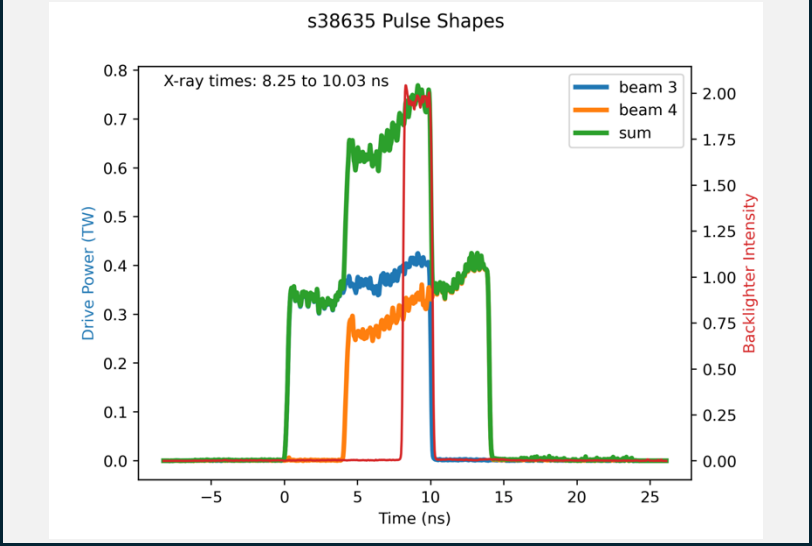
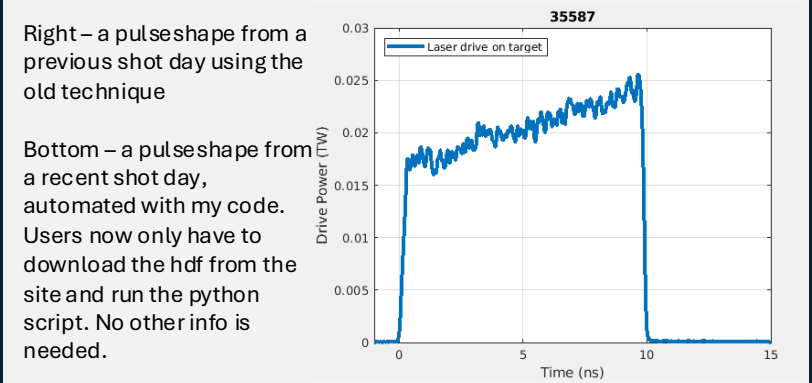
Other work @ JH

Automating Pulseshape Analysis

One of the essential functions at laser shock experiments while experiments are going on is the pulse shape of the delivered drive to your sample. At Omega, this is uploaded via hdf file to their website. When I arrived, getting pulseshapes was a cumbersome process: 1. download HDF, 2. Use HDFViewer to open HDF, 3. copy data into matlab, 4. deduce which beam is which, 5. plot and label. The whole process took ~20-30 min. I automated this to take ~1 min.

Right – a pulseshape from a previous shot day using the old technique

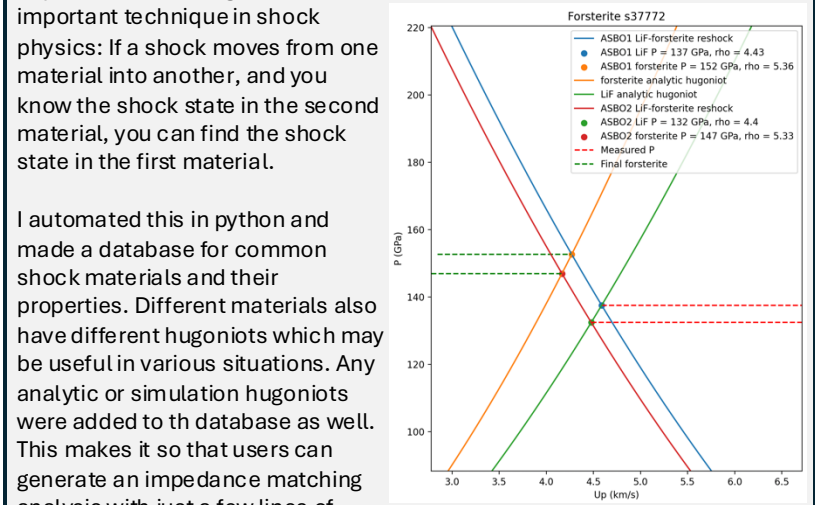
Bottom – a pulseshape from a recent shot day, automated with my code. Users now only have to download the hdf from the site and run the python script. No other info is needed.



Automating impedance matching

Impedance matching is an important technique in shock physics: If a shock moves from one material into another, and you know the shock state in the second material, you can find the shock state in the first material.

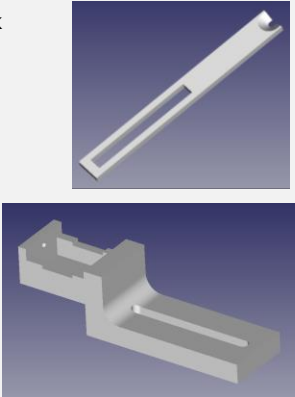
I automated this in python and made a database for common shock materials and their properties. Different materials also have different hugoniot which may be useful in various situations. Any analytic or simulation hugoniot were added to the database as well. This makes it so that users can generate an impedance matching analysis with just a few lines of code.



Experiment Support & Mentoring

Not everything has a nice plot to show. Here are some less tangible contributions and accomplishments I had:

- Mentored 2 interns in Summer 2024
 - Each intern carried out a shock physics project
 - I've continued working with one intern; she is presenting her work on Forsterite at the 2025 APS shock conference in June 2025
- Experiment support
 - 6x at University of Rochester LLE Omega
 - Live data analysis
 - Texas Petawatt laser
 - 1 month expt
 - Aligned optics
 - Soldered motors
 - Operated vacuum
 - Designed & printed replacement parts



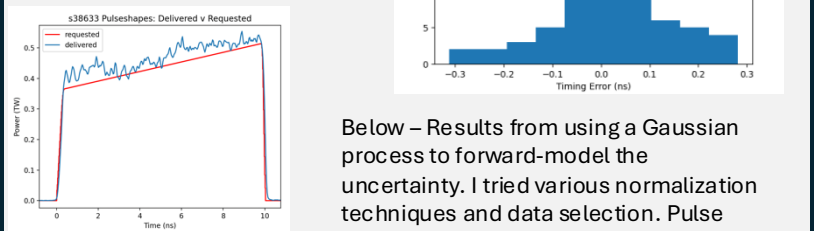
Below – Sample stock and holder I designed and 3d printed for the Texas Petawatt experiment.

Omega Pulseshape Uncertainty

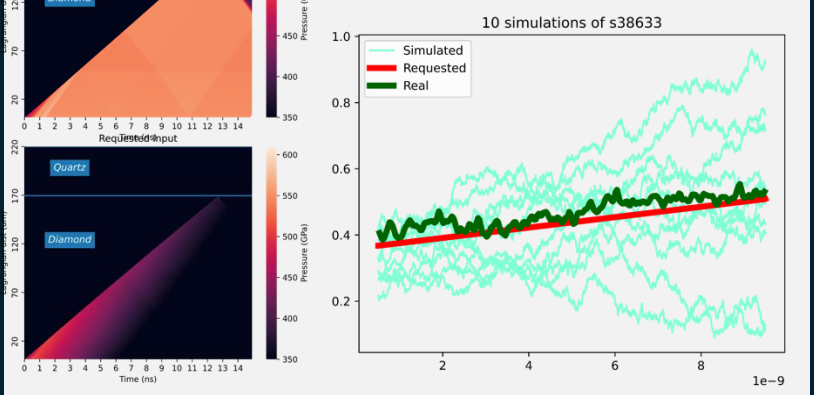
Laser pulse shaping is an integral part of many shock experiments. This is a complicated process and often the pulse shape requested by a user has a different power from the pulse delivered by the facility (as seen in the top plot at right). It is desirable to predict and simulate different pulseshapes which might be delivered.

Below (3 stacked plots) – The effect of a different pulseshape. The top plot shows delivered (blue) vs requested (red) pulses. The actual pulse was higher by ~200 GPa.

Right bottom – Timing differences between requested and delivered pulses.



Below – Results from using a Gaussian process to forward-model the uncertainty. I tried various normalization techniques and data selection. Pulse shaping at Omega is constantly iterating and improving, which makes data unusable if it is too old. Ultimately, I was not able to develop a useful model.



Anchor Bot: Daring Engineering

Project Requirements

- Create robot to install helical anchors on the sea floor with a certain uplift capacity
- DOE-funded kelp bed project

My Contributions

- Modeled robot requirements as a function of: Anchor uplift capacity, soil properties, propeller type, & motor power
- Determined soil properties of test site
- Designed and iterated mechanism of attaching the robot to the anchor
- Designed battery housing
- Performed Analysis of alternatives for and selected: motors, batteries, propellers

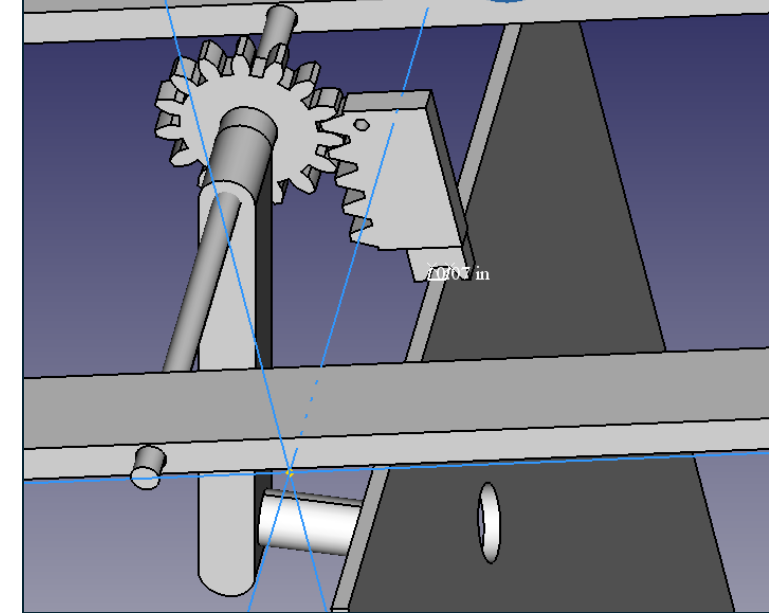
Outcomes

Robot was produced after ~ 6 months and successfully installed an anchor on the sea floor. Last-minute change to uplift capacity meant the robot lacked the energy for repeat installations

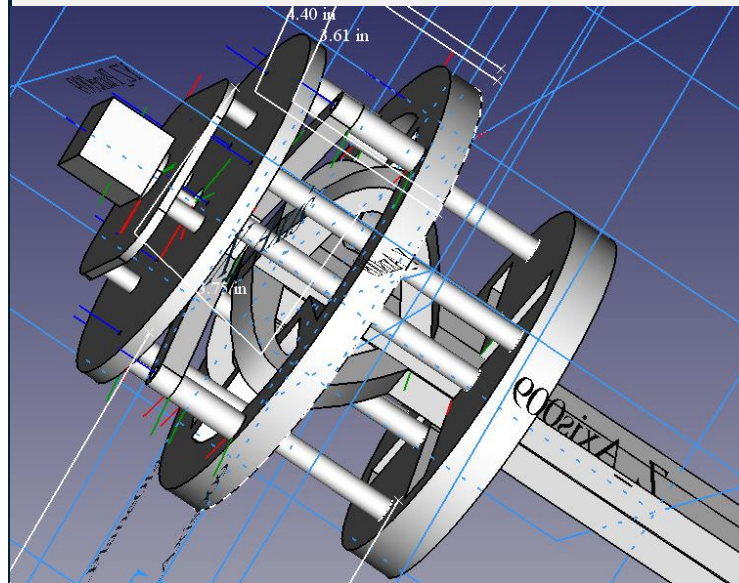
Task Overview

- Anchors on the sea floor kelp farms
- Anchors need a certain resistance to upward force
- Resistance to upward force is proportional to net torque during installation (also depends on anchor shape and soil properties)
- We made assumptions about environmental factors and designed and tested a robot that met the requirements

Anchor attachment design 1



Anchor Attachment Design 2



Battery Housing Design

